ISYE 6740 – Fall 2025
Final Report

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Project Title: NYC Parking Violations Predictions and Risk Warning

Problem Statement

New York City is a highly dense and populated city with millions of residents and visitors each day. On average, around 1.5 million cars enter New York City making it a hotspot for traffic incidents and parking violations. Public datasets reveal strong spatial and temporal patterns in enforcement: specific precincts and corridors issue disproportionately high numbers of tickets, and fines cluster around school zones, bus corridors, and commercial districts during weekday business hours. The tightly packed streets also make parking difficult, leading to people parking illegally and getting violations. Therefore, there are certain areas and pockets that can lead to more or less violation activity simply due to the nature of the street and the specific rules. Additionally, there are certain types of drivers and vehicles that can be more prone to violations. For example, out of state drivers, rental cars and electric vehicle drivers are all more prone to fines and violations since they are unfamiliar with New York City rules and regulations. As with most data, time is also an important factor in determining events. With this data, the time of day, time of the week or month (weekends, holidays etc.) can have a major impact on the number and severity of the violations.

Therefore, the goal of this project is to determine the areas where these violations frequently occur and use other characteristics to evaluate the risk factors for parking in certain areas. The models will take all factors into consideration including time, driver personas, violation intensity, fine amount, parking location etc.

The city issues millions of violations annually, generating hundreds of millions of dollars in fines and causing constant frustration among visitors and daily commuters to the city. Although traffic violation data is openly available, there is currently no public-facing system that warns travelers about the likelihood and potential severity of receiving a ticket at a specific destination. Additionally, drivers lack visibility into which areas are high risk, what types of violations are most common there, and how factors like vehicle registration state, time of day, or vehicle type affect their exposure.

Hence, another goal of this project is to utilize the data and predictions made about the violations to produce a risk and warning system that tells drivers the risk of parking in a certain area.

Data Source

I will be using Parking Violations Issued [1] from NYC Open Data from the NYC city government. The data includes parking violations from July 1, 2023 till November 2025 and includes fields such as plate state, vehicle make, issue date/time, precinct, violation code, and description.There are 43 columns and 35.6 million rows in the dataset, making it a sizable population to conduct analysis on. For the purpose of this analysis, I will only consider 6 months of data, from May 2025 to October 2025 (1,840,000 rows). The primary unique identifier is the Summons Number given on the violation/ticket. This dataset represents a frozen annual snapshot at the time of issuance and does not include later payment or dismissal updates.

Methodology

The first step was exploratory data analysis to clean and normalize and also understand the data. Given the data, I made sure to remove and supplement the null or unknown values with relevant substitutions. Then, I was able to view the data for initial patterns that could shape the overall results. The main data points I inspected were the number of violations per state, violations per county and the number of violations, for the top 5 violations. This was helpful to how the data was distributed and as expected, most of the violations were from New York state cars. There was also a decent spread of violations across the boroughs, mostly relative to their sizes and the number one violation as expected was speeding, followed by bus lane violations.

State	# of Violations
NY	1,319,020
NJ	167,704
PA	96,223
FL	42,598
CT	38,097

County (Borough)	# of Violations
Queens (QN)	559,106
Brooklyn (BK / Kings)	467,958
Bronx (BX)	230,496
Manhattan (MN)	193,014

Violation Description	# of Violations
PHTO SCHOOL ZN SPEED VIOLATION	908 ,449
BUS LANE VIOLATION	450 ,904
NO PARKING – STREET CLEANING	278 ,102
FIRE HYDRANT	37 ,564
NO STANDING – DAY/TIME LIMITS	26 ,964

Table 1, 2, 3: Exploratory Data Analysis on violations across states, boroughs & violation types

I also mapped out the Violations according to time such as day of the week, month and time of day. The results for day of the week and month were standard and showed a fair spread across each day and month. The time of day when violations occurred showed more variation and as seen in the chart below, peaked between the hours of 8 AM to 12 PM and slowed down for the remainder of the day into the early hours.

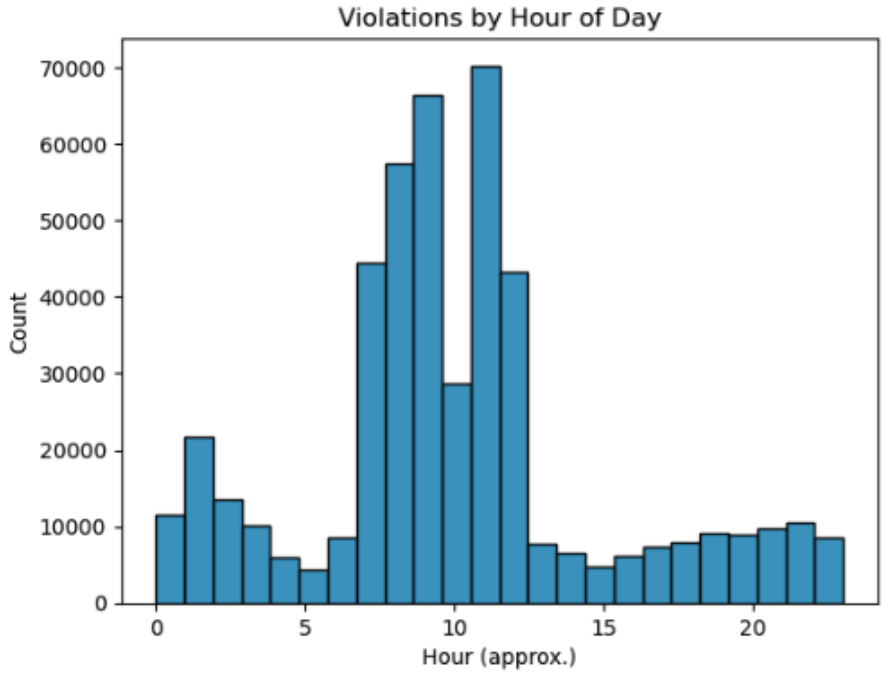


Figure 1: Vertical bar chart to show spread of violations throughout the day

I also explored the number of violations per precinct and the data here was standard with some precincts having more violations than others. This insight will be used later when connecting this violation data to GeoSpatial data to determine the highest risk areas.

Another preliminary data point I wanted to explore was the fine amounts. The fine amounts can indicate the severity of the violations and the intensity of certain violations over others. This data point is essential in building a risk scale. Here, we can see how the Fine (base cost of the violation) and Penalty (additional late fee if penalty is not paid) amounts are distributed:

Statistic	Fine Amount ($)	Penalty Amount ($)	Total Due ($)
Count	473,370	473,370	473,370
Mean	78.69	13.25	91.94
Standard Deviation	24.02	18.55	31.19
Minimum	0	0	0
25th Percentile (Q1)	65	0	65
Median (Q2)	65	10	75
75th Percentile (Q3)	95	10	125
Maximum	500	60	530

Table 4: Exploratory Data Analysis on fine amount and penalty amount

Here, we see that the mean fine amount is around $78.69 and the mean penalty amount is around $13.25. This helps to understand the severity and intensity of most of the tickets that people receive. When plotted, it is even easier to visualize and we can see the majority of the violations are in the range of $70 to $120 with more other higher range amounts as well.

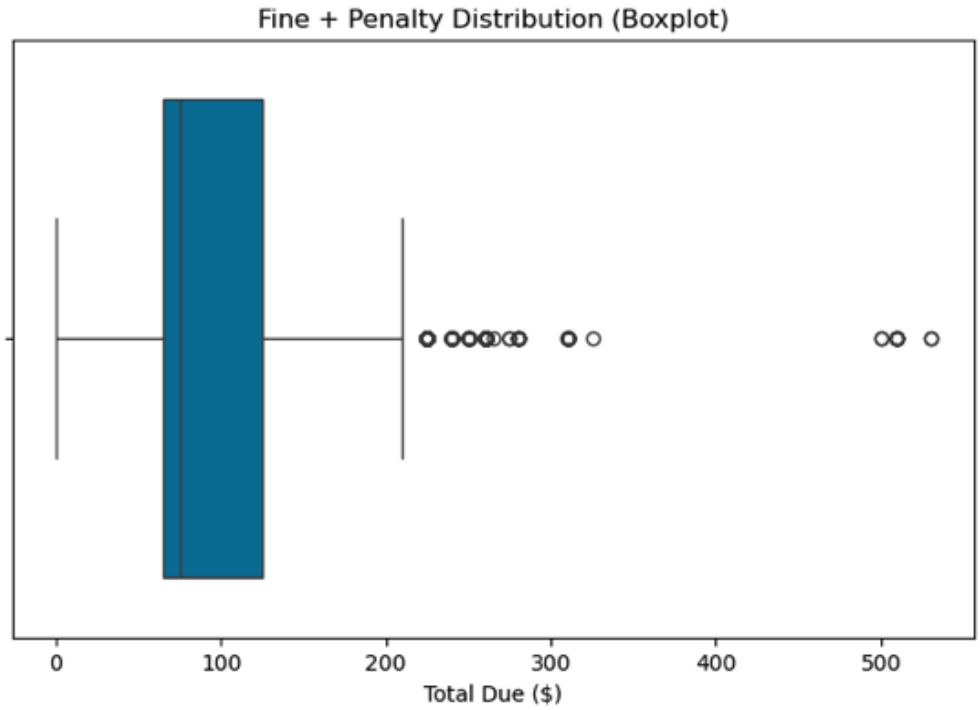


Figure 2: Box plot on spread of fine and penalty amount distributions

After completing exploratory data analysis to clean and normalize the data, I conducted feature engineering to extrapolate and aggregate the key information needed for the logistic regression analysis. Feature engineering is critical in the process because the raw parking violation dataset mainly provides transaction-level ticket data such as date, time, location, plate. To predict violation severity and risk, the data needs to be summarized and structured into key features like:

1. Camera Score — density/proximity of enforcement cameras
    - **What**: Quantifies how heavily monitored a location is with traffic cameras.
    - **How**: Aggregates number of cameras in a radius
    - **Insight**: areas with more cameras indicate areas with more law enforcement and higher chances of getting a ticket.

2. Violation Density Score — historical ticket frequency
    - **What**: Quantifies how frequently tickets are given in that area
    - **How**: Aggregates violations by precinct area
    - **Insight**: areas that historically have more violations are more likely to give out violations and being more risky

3. Temporal Features — weekday/peak hours

    - **What**: time and model reinforcement patterns over time, seasonality
    - **How**: Compares peak hours, weekends and day and night ratios
    - **Insight**: seasonality and trends could show that there is a spike in tickets during weekday rush hours and business days

4. Driver Persona Features — state, rental flag, EV status
    - **What**: Quantifies how driver persona traits correlate to higher violation rates
    - **How**: Compares state registrations, license plate types, electric vehicle status etc to measure violation severity
    - **Insight**: Certain driver personas (e.g. rental car drivers) may have higher chances of getting violations

5. Spatial Units - Precinct (County & Precinct)
    - **What**: What area that violation actually occurred
    - **How**: Overlays violation locations with zones such as commercial, residential, school zone etc.
    - **Insight**: Certain zones (e.g. school zone) may have higher severity and likelihood of violations

6. Severity (Fine and Penalty Amounts)
    - **What**: Quantifies how severity of fine and penalty amounts
    - **How**: Compared fine and penalty amounts in different areas
    - **Insight**: Higher fines and penalty amounts indicate higher risk and more law enforcement.

These aggregated features were used in calculating the R Scores to determine the risk severity and building the logistic regression model to predict the severity and probability of receiving a ticket.

Evaluation and Results

Then, to determine the risk severity, I calculated the R-score using a composite risk index. The weighted features are used to predict the likelihood of receiving a violation in different areas in NYC. Since there are multiple features, they need to be given weights that puts appropriate emphasis on intensity and how it contributes to revving a ticket.

Here the weight heuristic is defined as:

$$Ri = 0.45(camera) + 0.35(density) + 0.15(time) + persona\ adjustment$$

Therefore, the presence of cameras in the area is given the most importance and violation density is the next most important. Other factors like time and driver personas are not as crucial and therefore the weight is adjusted accordingly. The weights are tuned iteratively and adjusted to ensure that the R is positively correlated with the violation severity. Here, we can see there is a positive and upwards correlation to the R and the is high severity score indicating that higher R means a higher severity:

Variable	R	Is High Severity	Density Score	Camera Score	Time Score
R	1	0.138909	0.659724	0.546584	0.495283
is_high_severity	0.138909	1	0.085043	0.038653	0.144308
density_score	0.659724	0.085043	1	-0.074242	0.10334
camera_score	0.546584	0.038653	-0.074242	1	0.049831
time_score	0.495283	0.144308	0.10334	0.049831	1

Table 5: R Score for selected features to calculate risk severity

The results below show a Risk Tier chart by County/Precinct, the evaluated R score, the corresponding risk tier (High/Medium/Low) and the top 3 violations in that area. As the R score rises, the risk intensity increases and based on this scale, the tier is assigned:

High ($R \geq 0.70$), Medium ($0.40 \leq R < 0.70$), or Low ($R < 0.40$)

County	Precint	Day	R Today	Risk Tier	Top 3 Violations
BX	49	2025-10-31	0.882576	High	FIRE HYDRANT, OBSTRUCTING DRIVEWAY, NO STANDIN...
BX	46	2025-10-29	0.868154	High	DOUBLE PARKING, FIRE HYDRANT, OBSTRUCTING DRIV...
NY	25	2025-10-31	0.837808	High	NO PARKING-STREET CLEANING, SIDEWALK, NO STAND...
K	83	2025-10-31	0.835523	High	OBSTRUCTING DRIVEWAY, FIRE HYDRANT, DOUBLE PAR...
NY	34	2025-10-31	0.720991	High	NO PARKING-STREET CLEANING, DOUBLE PARKING, FI...
BX	44	2025-10-29	0.709683	High	DOUBLE PARKING, FIRE HYDRANT, NO PARKING-DAY/T...
BX	48	2025-10-31	0.684307	Medium	FIRE HYDRANT, DOUBLE PARKING, FRONT OR BACK PL...
Q	103	2025-10-31	0.675316	Medium	OBSTRUCTING DRIVEWAY, NO STANDING-DAY/TIME LIM...
K	5	2025-10-29	0.6	Medium	FIRE HYDRANT, NO PARKING-DAY/TIME LIMITS, NO P...
NY	14	2025-10-30	0.592778	Medium	NO STANDING-DAY/TIME LIMITS, NO STANDING-EXC. ...

Table 6: Risk tiering results sample based on R Scores and Top 3 violations.

Apart from the primary factors being derived from the main NYC Parking Violations dataset, the data was also integrated with data from geospatial boundary files of NYC Police Precincts [2]. The geospatial boundary files allow mapping of the data over a detailed map and results in the below visual. This is a warning system that utilizes colors and intensity to indicate the risk, with darker colors indicating higher severity.

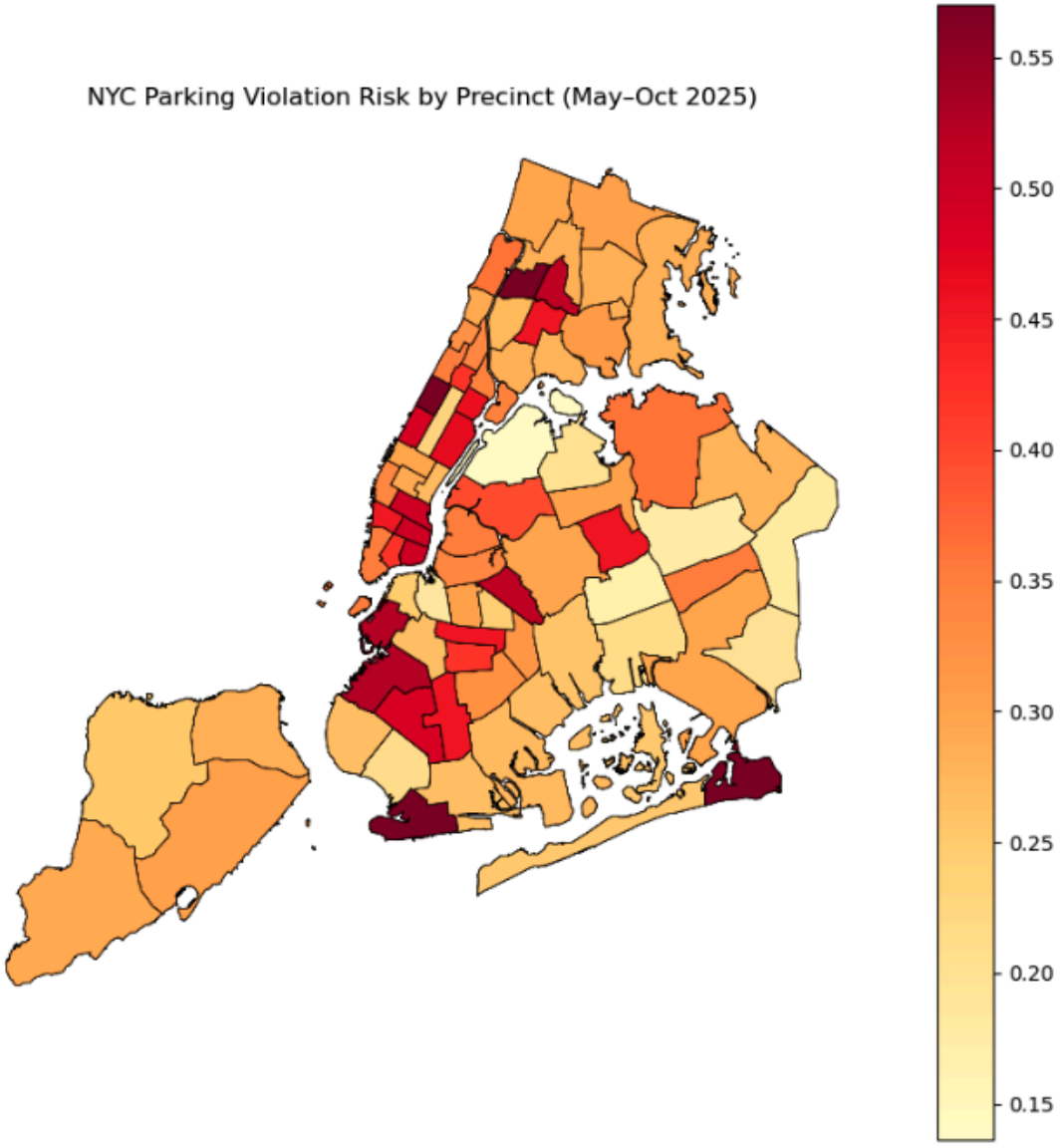


Figure 3: Risk severity mapped over NYC police precincts to show risk of violation in the area

Another component in this process is a logistic regression model that uses the historical tickets to translate the R-score into a probability of receiving a high-severity violation. Logistic regression will allow us to calibrate the R score and combine it with the other calculated features and personas to determine the violation levels. This will provide not just the severity based on historical data but also the probability of receiving a violation in certain areas, currently.

In the results, we see that the R Score produced a positive coefficient, which again confirms the idea that higher risk scores correlates to a higher likelihood of receiving a ticket.

Coefficient: 3.614265572691157
Intercept: -4.036070571925707

The probability is then defined as where 3.6143 is the coefficient (how much the probability is increased for an increase in the r score) and -4.0361 is the intercept (baseline probability when R score is 0). Given that an R score of 0 means low severity, it reasonable that the intercept is negative, meaning that the areas that have very low R score have a negative and very low likelihood of receiving a ticket.

$$P(\text{high severity}) = \sigma(-4.0361 + 3.6143 \cdot R)$$

Additionally, we can look at the classification results for this model to determine if the model correctly identifies high and low severity violations. The results are below:

	precision	recall	f1-score	support
0	0.96	1.00	0.98	135675
1	0.00	0.00	0.00	6336
accuracy			0.96	142011
macro avg	0.48	0.50	0.49	142011
weighted avg	0.91	0.96	0.93	142011

ROC AUC: 0.6982656029848885

Table 7: Model classification results

Here, we have the results for Class 0 (Low Severity) and Class 1 (High Severity) and their corresponding statistics. However, this indicates that while the model classifies very accurately on Class 0 (Low severity) violations, it does poorly at classifying Class 1 (High severity) violations. With the support data, we can see that out of the total of 142,011, only 6,336 are Class 1 violations. This is around 4.4% of the total data and shows that the data is imbalanced in the split across high and low severity violations.

Still the ROC AUC (Area Under the Curve of the Receiver Operating Characteristic) is around 0.6982 which indicates that the model does show some accurate predictions rather than just random guessing. Plotting the ROC curve with the true positive and false positive rates, results in the below:

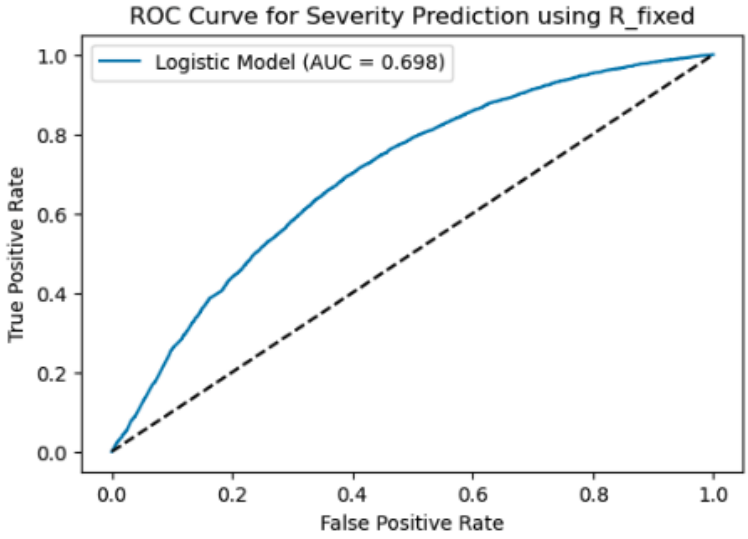


Figure 4: ROC Curve for severity prediction

Conclusion

In conclusion, this project leveraged NYC Parking Violation open data to build an interpretable and data-driven system for identifying risk severity and predicting parking violation risk. By combining heuristic scoring with logistic regression, the model highlights spatial, temporal, and driver-specific risk factors, while the warning system mapped with the geospatial results makes these insights actionable for travelers. The risk tiers and likely violation types for any given location are essential in helping drivers make smarter parking decisions and potentially reducing unnecessary fines in high-enforcement areas.

References

[1] URL: https://data.cityofnewyork.us/City-Government/Parking-Violations-Issued-Fiscal-Year-2024/pvqr-7yc4/about_data

[2] URL: https://hub.arcgis.com/datasets/c35786feb0ac4d1b964f41f874f151c1_0/about